How Holograms Work

01 What do ripples have to do with light?



Where ripples cross, bumps stack up or calm out.

Look closer. Find the tall bumps and the flat calm lines where the two rings cross.

A little explanation. Light travels in waves, a bit like ripples. Where two crests meet, they stack into a taller bump. Where a crest meets a dip, they cancel into calm. A hologram records that crossing pattern, made with light instead of water.

Try this. Put a little water in a baking pan. Tap two fingertips in it at the same time. Look for the calm lines between the bumps.

02 What is really on a hologram's film?



The film holds fine stripes, not a shell.

Look closer. Compare the shell with the fine stripes in the lens. Do they look alike?

A little explanation. Not a picture of the shell. A laser beam is split in two. One part bounces off the shell, and one part shines straight on the film. Where they cross, the film records light and dark stripes far thinner than a hair.

Try this. Look at your fingertip with a magnifying glass. Hologram stripes swirl a bit like that, but they are far tinier and closer together.

03 What happens if a hologram breaks?



Each half still shows the whole shell.

Look closer. Look inside each half. Is any part of the shell missing?

A little explanation. Each half still shows the whole shell, just a little blurrier. Light from every spot on the shell landed on every part of the film. It is like peeking through one pane of a window: you still see the whole yard.

Try this. Look out a window through a paper tube. Move the tube to a different pane. Can you still see the whole tree from each spot?

04 Are floating pictures in movies holograms?



Slanted glass makes a reflection seem to float.

Look closer. Find the real toy bird below and its reflection in the slanted glass.

A little explanation. Usually not. Many stage holograms are an old trick from 1862 called Pepper's ghost. A slanted sheet of clear glass reflects a lit figure, so it seems to float. A real hologram needs film or foil to bend the light.

Try this. At night, look out a dark window from a bright room. Can you see your room floating outside? That is the same glass trick.